Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials

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Abstract

Background. Aggregated N-of-1 trials can test whether a baseline biomarker predicts the response to a drug. Simulation studies of that test encode the biomarker interaction as a continuous effect. The interaction can instead be a latent-class structure, in which the biomarker predicts membership in a responder class. We ask whether a class-aware analysis has any advantage over a linear mixed model when the class structure is real.

Methods. We define the continuous and latent-class encodings and give the marginal correlation induced by a two-class mixture. We report a 240-replicate pilot and two 2,000-replicate null cells at $N = 35$ and $N = 70$, all on an open-label design. We describe a corrected

study on the Hybrid design at $N \in \{35, 70, 140\}$ that compares a linear mixed model with class-aware tests under identical fixed effects. The class-aware tests include a single-resample parametric-bootstrap likelihood-ratio test of two classes against one. The study is exploratory.

Results. The pilot's open-label design had no off-drug visit. Its linear mixed model therefore fitted a biomarker-by-time term, not the biomarker-by-drug interaction. That test rejected 0 of 240 null replicates and 5 of 2,000 and 1 of 2,000 in the production null cells. Its model standard error exceeded the empirical standard deviation by a factor of 1.67. The unconditional Wald test on the gating slope of a two-class mixed model rejected 0.197 of null replicates in the pilot (MCSE 0.027). It rejected 0.166 and 0.240 in the production cells. The gating slope exists only when two classes exist, so the Wald statistic has no chi-square limit under one class. BIC selected two classes in 0.4 to 5.1 percent of pilot fits. The corrected study's results will be inserted when the run completes.

Conclusions. The pilot and the production null cells do not answer the question. They show that an unconditional gating Wald test is anti-conservative at every sample size examined, and that a BIC gate turns it into a near-never-reject rule. The corrected study tests calibrated procedures on a design with an off-drug contrast.

Trial registration. Not applicable.

Keywords: N-of-1 trials; predictive biomarker; latent class; finite mixture; growth mixture model; simulation; Type I error.

1 Background

1.1 Aggregated N-of-1 trials and predictive biomarkers

In an aggregated N-of-1 trial each participant moves between drug and placebo, or between drug and no drug, several times. The individual series are analyzed together (1–5). Every participant contributes an on-drug and an off-drug response, so the design can test whether a baseline measurement predicts who benefits (6,7). The statistical target is the biomarker-by-treatment interaction $\beta_{bm:D}$ in a within-participant outcome model. Its power depends on the strength of the interaction, on the precision of the within-participant contrast, on the design, and on carryover between phases. Power is estimated by simulation (8,9). Hendrickson et al. (10) proposed a simulation method for the Hybrid design, which combines open-label titration, blinded discontinuation and a brief crossover. We call their process the original DGP.

1.2 Two encodings of a continuous interaction

The original DGP encodes the interaction as a correlation. The baseline biomarker and the drug-response component at every visit are drawn from one multivariate normal distribution. The predictive biomarker is a correlation c_{bm} between the biomarker and the drug response that is present on drug and absent off drug. We call this the correlation encoding. The

alternative in the trial literature shifts the drug-response mean by the centered biomarker, so the interaction is a coefficient β_{bm} in the conditional mean. We call this the mean-shift encoding. The companion paper (11) compares the two. At a matched total $N = 70$ in the Hybrid design, carryover with a half-life of one week reduced power by 15.4 percent under the correlation encoding and by 6.9 percent under the mean-shift encoding. Conditional on the biomarker, the correlation encoding is a mean shift with a smaller residual variance, so the two differ in the exposure profile of the interaction, not in kind.

1.3 The latent-class encoding

Both encodings are continuous. Every participant has some degree of drug response, and the response is a smooth function of the biomarker. A third encoding is discrete. Participants belong to one of a small number of unobserved classes, most often a responder and a non-responder class. The classes have distinct drug-response curves. The biomarker predicts class membership without determining it. The marginal interaction then arises from the mixture, not from a conditional-mean function. The discrete encoding corresponds to a different biological hypothesis. A metabolizing or non-metabolizing allele, or a present or absent target receptor, produces classes. A receptor density or a baseline severity produces a continuous gradient. The two hypotheses can give the same biomarker-outcome association at a single visit and differ over a longitudinal series, because the class label is fixed within a participant while the within-class drug contrast decays under carryover.

1.4 Related work

The latent-class encoding has a direct precedent. Muthén et al. (12) fitted a growth mixture model to a randomized longitudinal trial and estimated the treatment effect within latent trajectory classes. Our encoding is that model with the biomarker as a class-membership covariate. The N-of-1 and crossover literature has also examined the responder dichotomy. Senn (13,14) showed that the patient-by-treatment interaction variance is identifiable only with repeated within-patient crossover, and that apparent responder classes are often artifacts of within-patient variation. Araujo, Julious and Senn (15) extended this by simulation. These papers address treatment-effect heterogeneity. They do not encode a biomarker-predicted class structure in a simulation DGP, which is the gap this paper addresses.

The finite-mixture machinery comes from psychometrics and econometrics. Latent structure analysis (16,17) and the finite-mixture monograph of McLachlan and Peel (18) supply the discrete model. Covariate-dependent gating of class membership was developed in marketing science (19,20) and in machine learning as mixtures of experts (21). The growth mixture model (22,23) and the heterogeneous random-effects model (24) place the mixture on the random effects of a longitudinal mixed model. Lubke and Neale (25) simulated the continuous-versus-discrete question directly, and Bauer and Curran (26) set out its conceptual problems. Growth and factor mixture models fitted to single-component non-normal data can extract spurious classes (27,28), and class enumeration at small N has limited power (29–31). The

`lcmm` package (32) and the joint latent class literature (33) implement these models in R. Reporting standards for mixture studies exist (34,35).

1.5 Aims

This paper has three aims. The first is to state the continuous and latent-class encodings of the biomarker interaction in one notation and to give the marginal correlation induced by a two-class mixture. The second is to report, for the record, a pilot and two production null cells on an open-label design, and to state what they do and do not show. The third is to describe and receive a corrected simulation on the Hybrid design that compares a linear mixed model with class-aware tests under identical fixed effects and with calibrated tests of class count. The paper does not derive identifiability conditions. It examines by simulation whether the gating slope of a two-class model is testable at $N \in \{35, 70, 140\}$. The study is exploratory.

2 Methods

2.1 Notation

B_i is the pre-treatment biomarker of participant i and $b_i = (B_i - \bar{B})/s_B$ its standardized value. BR_{it} is the drug-response component of the symptom score Y_{it} at visit t. The original DGP adds a natural-history component TV_{it} and a placebo-response component PB_{it}. D_{it} is the binary drug state. $D_{bc,it}$ is the exposure-decayed drug indicator of the analysis model: 1 on drug, $\exp(-\lambda t_{sd,it})$ off drug after the first exposure, and 0 at off-drug visits before the first exposure. Here $t_{sd,it}$ is the time since discontinuation and $\lambda = \ln 2/t_{1/2}$ with $t_{1/2}$ the carryover half-life. $D_{e,t}$, written `De` in code, is the design expectancy η_t: 1 in the open-label phase, 0.5 when blinded and 0 at baseline. Z_i is the unobserved class label. ω is a mixing proportion and $\omega(B_i) = \Pr(Z_i = 1 \mid B_i)$ the class-membership probability. ψ_g is the gating slope, the steepness of $\omega(B_i)$ in b_i; the code calls it `gs`. $\mathbb{E}[\cdot]$ is the expectation. κ is the mean model standard error divided by the empirical standard deviation of the estimates. MCSE is the Monte Carlo standard error $\sqrt{\hat{\pi}(1-\hat{\pi})/n}$ of a rejection rate $\hat{\pi}$ over n replicates. The symbol ψ_g is unrelated to the γ of the companion component-decomposition paper (36).

2.2 The two-class mixture DGP

If the biomarker indexes an unobserved responder dichotomy, the generative form is a two-component mixture,

$$Z_i \mid B_i \sim \text{Bernoulli}(\omega(B_i)), \qquad BR_{it} \mid Z_i = z \sim f_z(t, D_{bc,it}),$$

with a logistic gate in the standardized biomarker,

$$\omega(B_i) = \text{expit}(\psi_0 + \psi_g\, b_i).$$

In every simulation below $\psi_0 = 0$, so $\omega(\bar{B}) = 0.5$. The class-conditional distributions f_0 and f_1 differ in their drug-response parameters. Two consequences follow. The class label is fixed across all visits of a participant. The class-conditional mean still depends on $D_{bc,it}$, so the two class means converge when $D_{bc,it} \to 0$ after a long off-drug period. An analysis with access to Z_i could use the fixed label where the instantaneous mean separation has decayed. The correlation encoding has no such label. Its correlation is the whole signal, and carryover erodes it directly.

2.3 The MVN approximation

A two-component mixture with class-dependent means and a common within-class covariance has a marginal distribution of (B_i, BR_{it}) whose first two moments are reproduced by one multivariate normal with $\text{Cor}(B, BR) = c_{bm}$. Suppose B_i and BR_{it} are independent given Z_i. Then $\text{Cov}(B, BR) = \omega(1-\omega)\Delta_B\Delta_{BR}$ with $\Delta_B = \mu_B^1 - \mu_B^0$ and $\Delta_{BR} = \mu_{BR}^1 - \mu_{BR}^0$, and

$$c_{bm}^2 \;=\; \frac{\omega(1-\omega)\Delta_B^2}{\text{Var}(B)} \;\cdot\; \frac{\omega(1-\omega)\Delta_{BR}^2}{\text{Var}(BR)}.$$

The squared marginal correlation is the product of the between-class variance fractions of B and of BR. The marginal correlation therefore equals the geometric mean of the two between-class variance fractions. If either class separation is small, the marginal c_{bm} is small even when the classes are well defined. The formula conditions on Z_i. It requires $B_i \perp BR_{it} \mid Z_i$, which is a restriction beyond the mixture DGP above, where B_i enters only through the gate. Within that restriction, and with ω bounded away from 0 and 1, a single multivariate normal with a correlation that is present on drug and absent off drug reproduces the mixture's second moments. A power calculation for a Wald test of $\beta_{bm:D}$ in a linear mixed model then rests on those moments.

2.4 Where the approximation may fail

The approximation can mislead in three regimes. Under well separated classes the marginal distribution of BR is bimodal and no multivariate normal matches its tails. A mixture analysis might then exceed the power of a linear mixed model. We do not assert that it does. The pilot below shows the opposite at $N = 70$ on an open-label design, and the corrected study examines it on the Hybrid design. Under a steep gate the biomarker behaves as a class label and a thresholded mean-shift specification is the better approximation. Under class-varying covariance, if the residual variance or the serial correlation differs between classes, no single-component multivariate normal represents the generative structure. The class-scaling DGP below is in this regime.

2.5 The class-scaling DGP and the random-slopes DGP

The pilot used the DGP of `02-heterogeneous-slopes-dgp.R`, which we call the class-scaling DGP. It draws one trial from the original DGP at $c_{bm} = 0$, assigns each participant a class by the logistic gate, and multiplies the drug-response component at on-drug visits by 1 in class 1 and by 0 in class 0. The non-responder class therefore loses the drug-response mean and the drug-response variance alike. The class-conditional variances differ: the drug-response variance σ^2_{BR} is present in class 1 and absent in class 0 at every on-drug visit. The class-scaling DGP contains no random slope. The pilot's earlier label, "heterogeneous-random-slopes DGP", was wrong.

The random-slopes DGP of `07-true-heterogeneous-slopes-dgp.R` is the Study 3 DGP of the working protocol. Each participant's drug-effect slope is drawn from a class-conditional normal distribution, $\beta_R \sim \mathcal{N}(\mu_R, \sigma^2_R)$ in the responder class and $\beta_{NR} \sim \mathcal{N}(0, \sigma^2_{NR})$ in the non-responder class, with class membership from the same gate. This is the Verbeke-Lesaffre heterogeneous random-effects form (24). The corrected study runs both DGPs as distinct cells at the same gating slopes. In the working protocol the gating-slope cells belong to Study 3 and the single-component null cells to Study 5. The pilot's gating-slope cells were labeled Study 5 in the earlier draft; that label was wrong.

2.6 Testing for a second class

The pilot tested the gating slope with an unconditional Wald statistic from a two-class `hlme` fit. Under one class the gating slope is a parameter that exists only under the alternative. The information matrix is singular at the null, and the Wald statistic has no χ^2_1 limit at any sample size (37–39). The same problem affects the naive likelihood-ratio test of two classes against one, whose null distribution is not χ^2 on the parameter-count difference (18). The calibrated tests are the parametric bootstrap likelihood-ratio test (29,40), the Lo-Mendell-Rubin test (41), and the modified likelihood-ratio test of Chen, Chen and Kalbfleisch (42). Dziak, Lanza and Tan (30) give the power of the bootstrap test as a function of N and class separation at sample sizes that include the N-of-1 range. Local solutions are common when the multi-start budget is small (43). One-step estimation of a covariate in the class-membership model, as the `classmb = ~bm` specification does, is the standard approach, and its alternatives are the three-step methods of Vermunt (44) and Asparouhov and Muthén (45).

The earlier draft attributed the inflation of the gating Wald test to the overextraction mechanism of Bauer and Curran (27). That mechanism concerns spurious classes extracted from a single non-normal population. The null DGP here is a single multivariate normal, so the pilot does not test that mechanism. The inflation is the non-regular testing problem above.

The corrected study separates the two questions. It tests two classes against one with a single-resample parametric-bootstrap likelihood-ratio test. Under the null, one bootstrap sample from the one-class fit is drawn per replicate and the two fits are repeated on it. Across replicates the bootstrap statistics estimate the null distribution of the likelihood ratio, and each replicate's observed statistic is compared with that distribution. This is the single-

resample form of the bootstrap test. Conditional on two classes, the test of whether the biomarker predicts class membership is a regular problem and the Wald test on ψ_g applies. The BLRT-gated gating test rejects when both the bootstrap test and the gating Wald reject.

2.7 Analyses and test forms

The linear mixed model is $Y_{it} = \beta_0 + \beta_B B_i + \beta_T w_t + \beta_D D_{bc,it} + \beta_{bm:D} B_i D_{bc,it} + u_i + \varepsilon_{it}$, with a random intercept and a continuous-time AR(1) residual correlation, fitted by `nlme::lme`. The test of $\beta_{bm:D}$ is the Wald test on `bm:Dbc`, with model-based standard errors and, in the corrected study, also with CR2 cluster-robust standard errors and Satterthwaite degrees of freedom. The class-aware model is `lcmm::hlme` with the same fixed effects, a random intercept, class-specific coefficients on $D_{bc,it}$ (`mixture = ~Dbc`), and the biomarker in the class-membership model (`classmb = ~bm`). The one-class fit anchors the starting values of a `gridsearch` with five starts and 30 iterations for the two-class fit, followed by a final fit with 100 iterations. The class-aware test forms are the unconditional Wald on ψ_g, the within-class Wald on `bm:Dbc` from the two-class fit, the BIC-gated gating test, which rejects when BIC prefers two classes and the gating Wald rejects, and, in the corrected study, the BLRT-gated gating test. BIC selection of two classes is reported as an estimand in its own right.

On the open-label design of the pilot and of the production null cells, every post-baseline visit is on drug, so $D_{bc,it}$ equals 1 at every post-baseline visit and 0 at baseline. `lme_analysis()` detects that no participant has within-person variation in drug state and fits `Sx ~ bm + t + bm:t`. The reported coefficient is `bm:t`, a biomarker-by-linear-time term. The `hlme` wrapper fits `Sx ~ bm + t + Dbc + bm:Dbc`, in which `Dbc` is a post-baseline indicator. The two arms of the pilot therefore fitted different fixed effects, and neither tested $\beta_{bm:D}$ as defined above. The carryover half-life of 1.0 week set in the pilot had no effect, because $t_{sd,it} = 0$ at every visit.

2.8 Performance measures

For each cell we report the number of replicates, the fraction of two-class fits that converged (`conv = 1`), the rejection rate at $\alpha = 0.05$ with its MCSE for each test form over converged fits, the fraction of fits in which BIC preferred two classes, κ for the linear mixed model and for the gating coefficient, and the dispersion of the gating estimate as the median $|\hat{\psi}_g|$ and the fraction with $|\hat{\psi}_g| > 10$. Non-convergence is reported as an outcome. A test whose realized size is far below α is reported as conservative, not as controlled, because its power is not comparable with the power of a test at nominal size.

2.9 The pilot and the production null cells

The pilot used an open-label design with a baseline visit and eight post-baseline visits at 2.5-week intervals, all on drug, with one randomization path, constant expectancy and no blinding. The total sample size was $N = 70$. Three cells were run at gating slopes $\psi_g \in \{0, 1.0, 1.5\}$

under the class-scaling DGP, with 240 replicates each. The production null cells used the same design under the original DGP at $c_{bm} = 0$, at $N = 35$ and $N = 70$, with 2,000 replicates each. The 100-replicate pre-flight of the earlier draft used the same design at $N = 70$ under the null.

2.10 The corrected study

The corrected study uses the Hybrid design: open-label titration, blinded discontinuation, then a brief crossover, with a baseline visit and visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, and four randomization paths. The total sample size is $N \in \{35, 70, 140\}$ across paths, so $N = 140$ allocates about 35 per path. The DGP cells are the single-Gaussian null (the original DGP at $c_{bm} = 0$), the class-scaling DGP at $\psi_g \in \{1.0, 1.5\}$, the random-slopes DGP at the same gating slopes, and a correlation-encoding cell with c_{bm} matched to the marginal on-drug $\mathrm{Cor}(B, BR)$ of the mixture cells. The analyses have identical fixed effects: the linear mixed model with model-based and with CR2 inference, `hlme` with one and two classes, the unconditional gating Wald, the within-class `bm:Dbc` Wald, BIC selection, and the single-resample bootstrap likelihood-ratio test with the BLRT-gated gating test. The driver is `analysis/scripts/latent-class-mixture-application/12-referee-rerun.R`. It writes `summary.rds`, `calibration.rds` and `all.rds` to `analysis/data/latent-class-mixture-application/refer` The seed scheme follows the production driver: a fixed seed base plus the replicate index plus a cell offset, with `mc.set.seed = FALSE`. The seed base and the replicate count per cell are recorded in the driver and in `summary.rds`.

2.11 Reproducibility

The 240-replicate pilot was produced by `quick-sim-driver-parallel.R` in `analysis/scripts/latent-class-` on four cores with `parallel::mclapply`, wall time 558.6 s. Its seed for each replicate is `SEED_BASE + rep_idx + 1000 * cell_index` with `SEED_BASE = 20260507` and `mc.set.seed = FALSE`. Its per-replicate output is `analysis/data/quick-sim/03-latent-replicates.rds`. Four drivers in that directory write to the same output path, so the stored file is identified by its run log (`03-latent-runlog.txt`: four cores, 240 replicates per cell, 1,440 rows). The 100-replicate pre-flight was produced by `03-study5-pilot.R` with seed `20260507 + rep_idx` and is stored at `output/03-study5-pilot.rds` in the scripts directory. The production null cells were produced by `10-study5-production.R` with seed `20260509 + rep_idx + 1e6 * cell_id` and `mc.set.seed = FALSE`, and are stored at `analysis/data/study5/study5-progress.rds`, dated 2026-05-09. Every number in the Results is computed from these files when the manuscript is rendered. The working protocol is `00-ademp-protocol.md` and its deviations log `02-deviations-log.md` in the scripts directory.

3 Results

3.1 The pilot and the production null cells

We computed every entry of Tables 1 and 2 from the stored replicate files. These results are reported for the record. They are superseded by the corrected study, because the open-label design supplies no off-drug contrast and the two analysis arms fitted different fixed effects.

Table 1: Rejection rates at $\alpha = 0.05$ (MCSE in parentheses) over converged two-class fits, and the fraction of converged fits in which BIC preferred two classes (BIC two). The linear mixed model (LMM) column is the Wald test on `bm:t`, the term actually fitted on the open-label design, over all replicates. Conv. is the converged fraction. The pilot cells use the class-scaling DGP; the production cells use the original DGP at $c_{bm} = 0$.

Cell	N	Reps	Conv.	LMM Wald	Gating Wald	Within-class	BIC-gated	BIC two
Pilot, gating slope 0.0	70	240	0.929	0.000 (0.000)	0.197 (0.027)	0.000 (0.000)	0.004 (0.004)	0.004 (0.004)
Pilot, gating slope 1.0	70	240	0.988	0.325 (0.030)	0.304 (0.030)	0.430 (0.032)	0.025 (0.010)	0.051 (0.014)
Pilot, gating slope 1.5	70	240	0.983	0.542 (0.032)	0.360 (0.031)	0.521 (0.033)	0.004 (0.004)	0.017 (0.008)
Production null	35	2000	0.942	0.003 (0.001)	0.166 (0.009)	0.000 (0.000)	0.006 (0.002)	0.020 (0.003)
Production null	70	2000	0.947	0.001 (0.000)	0.240 (0.010)	0.000 (0.000)	0.006 (0.002)	0.009 (0.002)

Table 2: Calibration and dispersion diagnostics. κ is the mean model standard error divided by the empirical standard deviation of the estimates. Entries marked with an asterisk are computed from the stored p-value by normal inversion, because the production file does not store standard errors. The gating diagnostics use converged two-class fits.

Cell	N	κ, LMM	κ, gating	Median $\lvert\hat{\psi}_g\rvert$	Fraction $\lvert\hat{\psi}_g\rvert > 10$
Pilot, gating slope 0.0	70	1.67	13.5	1.43	0.179
Pilot, gating slope 1.0	70	1.79	7.8	0.99	0.072
Pilot, gating slope 1.5	70	1.70	12.5	1.14	0.093
Production null	35	1.83*	31.6*	2.19	0.266
Production null	70	1.75*	12.5*	1.39	0.176

The linear mixed model's test rejected 0 of 240 null replicates in the pilot. Under a test of size 0.05 that event has probability $0.95^{240} = 4.5 \times 10^{-6}$. The empirical standard deviation of $\hat{\beta}$ was 0.049 and the mean model standard error 0.082, so $\kappa = 1.67$. The smallest of the 240 null p-values was 0.090 and the median 0.69. The production null cells rejected 5 of 2,000 at $N = 35$ and 1 of 2,000 at $N = 70$. The test is severely conservative. Its power at gating slopes 1.0 and 1.5 is not comparable with the power of any test at nominal size, and the earlier draft's statement that it "holds nominal Type I" was wrong. The conservatism is a property of the `bm:t` specification with a CAR(1) residual on a Gompertz trajectory and is not diagnosed here.

The unconditional gating Wald rejected 0.197 of 223 converged null fits in the pilot and 0.166 (0.009) and 0.240 (0.010) in the production cells at $N = 35$ and $N = 70$. The inflation did

not shrink with sample size. The gating estimates were dispersed: the median $|\hat{\psi}_g|$ was 1.43 under the null and 0.179 of the estimates exceeded 10 in absolute value. These are boundary solutions with a near-empty class, the pattern that the non-regular testing problem predicts.

BIC preferred two classes in 0.4, 5.1 and 1.7 percent of converged pilot fits at gating slopes 0, 1.0 and 1.5, and a BIC difference above 6 never occurred. The BIC-gated gating test cannot reject more often than BIC selects two classes, so its rejection rates of 0.004 (0.004), 0.025 (0.010) and 0.004 (0.004) are bounded by the selection rate. The difference between the two non-null cells is within two MCSE and is not interpreted.

The within-class `bm:Dbc` Wald, the fourth recorded test form, rejected 0.000 (0.000) of null fits in the pilot and 0.000 (0.000) and 0.000 (0.000) in the production cells. At gating slope 1.5 it rejected 0.521 (0.033), within one MCSE of the linear mixed model's 0.542 (0.032). A test with a realized size of zero is uninformative about calibration, so this comparison does not rank the two tests.

The production file contradicts the earlier draft's statement that the production driver had not been run. The file is dated 2026-05-09 and holds 4,000 replicates in two cells, both under the single-Gaussian null, at $N = 35$ and $N = 70$. The driver's non-Gaussian and two-class cells are absent from it, so the run was incomplete.

3.2 The corrected study: Hybrid design, matched models, calibrated tests

The corrected study has not completed; its results will be inserted here when `summary.rds` is present.

Table 3: Corrected study: effect-size calibration

dgp_id	dgp	gating_slope	c_bm	n_draw	ref_visit	cor_bm_br	f2	var_br_all	mean_br_all
null	single-gaussian-null	NA	0.000	20000	OL2	0.004	0.000	64.301	9.259
cs-gs1.0	class-scaling	1.0	0.000	20000	OL2	0.254	0.069	53.023	4.634
cs-gs1.5	class-scaling	1.5	0.000	20000	OL2	0.334	0.126	52.797	4.569
rs-gs1.0	true-random-slopes	1.0	0.000	20000	OL2	0.242	0.062	67.855	4.607
rs-gs1.5	true-random-slopes	1.5	0.000	20000	OL2	0.309	0.105	65.402	4.581
mvn-cbm0.334	correlation-encoding-mvn	NA	0.334	20000	OL2	0.343	0.133	64.315	9.256

4 Discussion

The pilot and the production null cells establish two facts. An unconditional Wald test on the gating slope of a two-class mixed model is anti-conservative at $N = 35$ and $N = 70$, and the inflation does not decline with N. This is the expected behavior of a Wald test on a parameter that exists only under the alternative. A BIC gate removes the inflation by rarely firing. Neither fact bears on whether a class-aware analysis can compete with a linear mixed model when the class structure is real, because the open-label design supplies no off-drug contrast and the linear mixed model of the pilot tested a different term. The corrected study answers that question with matched fixed effects on the Hybrid design and with a bootstrap test of class count.

Three questions remain for the corrected study. The first is whether a class-aware test is usable as a primary analysis at any N-of-1 sample size, or only as a sensitivity analysis. The second is how the analysis half-life should be chosen when the DGP is a mixture, since class-conditional carryover may differ. The third is whether a design that maximizes the per-participant contrast gains power under a latent-class DGP that the Hybrid design surrenders.

4.1 Limitations

The stored results come from one design that carries no interaction information. The linear mixed model's conservatism on that design is reported but not diagnosed. The production run was incomplete, and its non-Gaussian and two-class cells were never produced. The corrected study uses a single-resample bootstrap test, which estimates the null distribution across replicates rather than within each replicate. The two-class `hlme` fits use a multi-start budget of five starts and 30 iterations, and the fraction of fits at the iteration cap is not stored. The application to the prazosin-PTSD data of the original study has not been carried out.

5 Conclusions

The latent-class encoding of a biomarker interaction is a growth mixture model with the biomarker as a class-membership covariate. Its marginal correlation equals the geometric mean of the between-class variance fractions of the biomarker and the drug response, so a single multivariate normal reproduces its second moments when the biomarker and the drug response are independent given class. Testing the gating slope without first establishing two classes is a non-regular problem, and the pilot's unconditional Wald test was anti-conservative at every sample size examined. The corrected study on the Hybrid design, with matched fixed effects and a bootstrap test of class count, is the comparison this paper set out to make.

Glossary

- **Gating slope** (ψ_g, `gs`): the coefficient of the standardized biomarker in the logistic class-membership model; the steepness of $\omega(B_i)$ in b_i.
- **`ng`**: the number of latent classes in an `lcmm::hlme` fit; one or two in this paper.
- **BIC gate**: the rule that the gating Wald test is performed only when the two-class fit has a lower BIC than the one-class fit; the stored quantity is `delta_bic`, BIC of the one-class fit minus BIC of the two-class fit, and the gate opens when it is positive.
- **BLRT gate**: the rule that the gating Wald test is performed only when the single-resample parametric-bootstrap likelihood-ratio test rejects one class in favor of two.
- **Single-resample bootstrap**: one bootstrap sample per replicate from the one-class fit, with the null distribution of the likelihood-ratio statistic estimated across replicates.
- **OL**: the open-label design, in which every post-baseline visit is on drug and there is one randomization path.

- **OL+BDC**: open label followed by blinded discontinuation, with two randomization paths.
- **Hybrid design** (`Hybrid`): open-label titration, blinded discontinuation, then a brief crossover, with visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20 and four randomization paths.
- `bm:Dbc`: the biomarker-by-exposure interaction term of the analysis model, whose coefficient is $\beta_{bm:D}$.
- **Exposure-decayed drug indicator** (D_{bc}, `Dbc`): 1 on drug, $\exp(-\lambda t_{sd})$ off drug after the first exposure, and 0 at off-drug visits before the first exposure.
- **Design expectancy** (D_e, `De`, η_t): the participant's probability of being on drug by design, 1 open label, 0.5 blinded and 0 at baseline.
- **Entropy** (`entropy`): one minus the mean over participants of the posterior class-membership entropy divided by ln(`ng`), as computed in the `hlme` wrapper; 1 means perfectly separated classes.
- **Class-scaling DGP** (`02-heterogeneous-slopes-dgp.R`): a trial drawn from the original DGP at $c_{bm} = 0$ in which the drug-response component at on-drug visits is multiplied by 1 in the responder class and by 0 in the non-responder class, with class membership from the logistic gate.
- **Random-slopes DGP** (`07-true-heterogeneous-slopes-dgp.R`): the Study 3 DGP, in which each participant's drug-effect slope is drawn from a class-conditional normal distribution, with class membership from the logistic gate.
- **Correlation encoding** (`mvn`): the interaction is a correlation c_{bm} between the biomarker and the drug response that is present on drug and absent off drug.
- **Mean-shift encoding** (`mean_moderation`): the interaction is a shift of the drug-response mean by the centered biomarker, with coefficient β_{bm}.
- **Original DGP**: the data-generating process of Hendrickson et al. (2020), with the correlation encoding.
- **Within-class Wald**: the Wald test on `bm:Dbc` from the two-class `hlme` fit, in which the coefficient is common to the classes.
- **Calibration ratio** (κ): the mean model standard error divided by the empirical standard deviation of the estimates; $\kappa > 1$ means a conservative test.
- **MCSE**: the Monte Carlo standard error of a rejection rate, $\sqrt{\hat{\pi}(1-\hat{\pi})/n}$ over n replicates.

Abbreviations

ADEMP: aims, data-generating mechanisms, estimands, methods, performance measures; BDC: blinded discontinuation; BIC: Bayesian information criterion; BLRT: bootstrap likelihood-ratio test; BR: drug-response component; CAR(1): continuous-time first-order autoregressive correlation; CR2: bias-reduced cluster-robust variance estimator; DGP: data-generating process; GMM: growth mixture model; LMM: linear mixed model; MCSE: Monte Carlo standard error; MVN: multivariate normal; OL: open label; OL+BDC: open label followed by blinded discontinuation; PB: placebo-response component; PTSD:

post-traumatic stress disorder; TV: natural-history component.

Declarations

Ethics approval and consent to participate. Not applicable: the study uses simulated data only.

Consent for publication. Not applicable.

Availability of data and materials. All code and simulated data are in the pmsimstats-ng repository (`https://github.com/rgt47/pmsimstats-ng`); a tagged release with a persistent identifier will be deposited at submission, and the commit reported here should be cited until then. The stored replicate files, drivers and seeds are listed in the Reproducibility section. The manuscript source, bibliography and CSL file are at `analysis/report/03-latent-class-mixture-application/`.

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